

My Study Notes

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1 Introduction

Welcome to my study notes. This document is organized into multiple chapters to keep everything clean and modular.

2 Conjectures and Theorems

2.1 Basic Definitions

Definition 1. A conjecture is a mathematical statement that is believed to be true based on evidence or intuition, but which has not yet been proved or disproved.

Definition 2. A theorem is a mathematical statement that has been rigorously proved to be true from previously established results (such as axioms, definitions, and other theorems).

2.2 Examples from Number Theory

Example 2.1 (A true conjecture that became a theorem). Historically, mathematicians conjectured that there are infinitely many prime numbers. This was later proved by Euclid and is now a theorem:

Theorem 1 (Euclid). There are infinitely many prime numbers.

Thus, a statement can start as a conjecture and, once proved, become a theorem.

Example 2.2 (A conjecture that is still open). Goldbach's Conjecture is one of the most famous unsolved problems in number theory:

Conjecture 1 (Goldbach). Every even integer $n \geq 4$ can be written as the sum of two prime numbers.

This statement has been checked for very large values of n , but no complete proof or counterexample is known.

Example 2.3 (A false conjecture). *Consider the following statement:*

Conjecture 2. *Every odd integer $n > 1$ is a prime number.*

At first glance, small examples like 3, 5, 7 might suggest this is true. However, it is false. For instance,

$$9 = 3 \cdot 3, \quad 15 = 3 \cdot 5,$$

so 9 and 15 are odd but not prime. These are counterexamples that disprove the conjecture.

2.3 Remarks

Remark 1. *Conjectures play a central role in mathematical research. They guide exploration and often lead to new techniques, whether they eventually become theorems or are disproved by counterexamples.*

3 Algebraic Identities

3.1 Introduction

Algebraic identities are formulas that allow us to rewrite expressions in a more useful or factored form. They play a central role in number theory, especially when studying divisibility, prime numbers, and the structure of integers.

In this chapter we explore classical identities, their applications, and historical curiosities.

3.2 Difference of Powers

One of the most important identities is the factorization of a difference of powers:

Theorem 2 (Difference of Powers). *For any integer x and positive integer n ,*

$$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \cdots + x + 1).$$

This identity is fundamental in the study of numbers of the form $x^n - 1$, especially when $x = 2$.

Example 3.1 (Factoring a Mersenne-Type Number). *Consider the number*

$$2^{15} - 1.$$

Since $15 = 3 \cdot 5$, we can apply the identity twice:

$$2^{15} - 1 = (2^3 - 1)(2^{12} + 2^9 + 2^6 + 2^3 + 1).$$

Then factor $2^3 - 1$ again:

$$2^3 - 1 = (2 - 1)(2^2 + 2 + 1) = 7.$$

This method reveals the internal structure of the number without guessing prime factors.

3.3 Sum of Powers

Another classical identity is the sum of powers:

Theorem 3 (Sum of Powers). *For any integer x and positive integer n ,*

$$x^n + 1 = \begin{cases} (x + 1)(x^{n-1} - x^{n-2} + \cdots - x + 1), & \text{if } n \text{ is odd} \\ \text{irreducible over the integers,} & \text{if } n \text{ is even.} \end{cases}$$

Example 3.2 (A Curious Identity). *Since 5 is odd,*

$$2^5 + 1 = 33 = (2 + 1)(2^4 - 2^3 + 2^2 - 2 + 1).$$

This identity explains why some numbers of the form $2^n + 1$ factor, while others (the Fermat numbers) are suspected to be prime.

3.4 Historical Notes

Remark 2 (Euclid and Mersenne). *Euclid already used the identity*

$$x^n - 1 = (x - 1)(x^{n-1} + \cdots + 1)$$

in his proof that there are infinitely many primes. Much later, Marin Mersenne studied numbers of the form $2^p - 1$, now called Mersenne numbers. Some of these are prime, and they produce the largest known primes in history.

Remark 3 (Fermat's Curiosity). *Pierre de Fermat conjectured that all numbers of the form*

$$F_n = 2^{2^n} + 1$$

were prime. He was correct for $n = 0, 1, 2, 3, 4$, but Euler disproved the conjecture by showing:

$$F_5 = 2^{32} + 1 = 4294967297 = 641 \cdot 6700417.$$

This is one of the most famous false conjectures in number theory.

3.5 Applications

- Factoring large integers using algebraic structure.
- Understanding prime candidates such as Mersenne and Fermat numbers.
- Simplifying expressions in modular arithmetic.
- Building conjectures about divisibility and primality.

3.6 Curiosities

- The largest known prime numbers are almost always of the form $2^p - 1$, where p is prime.
- The identity for $x^n - 1$ is used in cryptography, especially in algorithms involving cyclic groups.
- Some identities, like the sum of powers, behave differently depending on whether the exponent is odd or even — a subtle but important detail.